GENOMICS LABORATORY REPORT

Next-Generation Sequencing & Clinical Bioinformatic Evaluation
Accredited ISO-15189 / CLIA High-Complexity Laboratory

ID: REP-2026-0005
Status: DRAFT
Date: Oct 1, 2026

1. REQUISITION & INVESTIGATOR INFORMATION

Requester Name:	Dr. Priya Nair	**Requisition Title:**	Live Verification - Targeted Liquid Biopsy Panel
Organization / Dept:	Institute of Rare Genetic Diseases	**Sample Type:**	Blood
Request Code:	REQ-2026-0012	**Priority Level:**	URGENT

2. SPECIMEN IDENTIFICATION & QUALITY METRICS

Sample Barcode:	SMP-2026-0014
Collection Date:	1/10/2026
Received Date:	1/10/2026

QC PARAMETER	MEASURED VALUE	QC STATUS
Fluorometric Conc.:	48.7 ng/µL	**PASS**
Purity A260/280 & 230:	1.84 / 2.15	RIN/DIN: 9.2

3. SEQUENCING RUN SPECIFICATIONS

Run Code:	RUN-2026-0001	**Run Assay Type:**	WGS
Platform / Chemistry:	Illumina NovaSeq	**Read Configuration:**	2x150bp
Target Coverage:	30x target depth	**Execution Status:**	COMPLETED

4. BIOINFORMATICS PIPELINE & INTERPRETATION SUMMARY

Pipeline: BWA-MEM + GATK HaplotypeCaller + Mutect2

Reference Assembly: GRCh38

Executive Findings & Summary:

High-confidence detection of KRAS p.Gly12Asp mutation suitable for targeted clinical intervention.

5. LABORATORY CERTIFICATION & SIGN-OFF

This test was developed and its performance characteristics determined by the Genomics Laboratory. It has been validated in accordance with laboratory accreditation standards for research and clinical oversight.

VERIFIED BY (LAB STAFF):
Sarah Lin, M.S.
Date: 2026-10-01

APPROVED BY (LAB DIRECTOR):
Laboratory Medical Director, Ph.D., FACMG
Electronic Signature Verification: PASSED